

IoT-based Industrial SCADA System for Real-Time Monitoring and Automated Alerts

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Abstract— This project presents a microcontroller-driven IoT-based SCADA (Supervisory Control and Data Acquisition) through real-time monitoring. The system tracks critical environmental and operational parameters such as gas concentration, temperature, humidity, and vibrations. It uses a range of sensors: MQ7 for carbon monoxide, MQ2 for various flammable gases, DHT11 for temperature and humidity, and vibration sensors to detect unusual physical movements. Sensor data is collected via a NodeMCU microcontroller and transmitted wirelessly to the ThingSpeak cloud platform, where it is logged and visualized for continuous remote monitoring. In the event of any abnormal condition such as elevated gas levels, overheating, or unusual vibrations the system triggers immediate alerts. These alerts are issued in two ways: audible warnings via buzzers and SMS notifications through a GSM module. Additionally, an LCD display provides real-time, on-site data visibility. The entire setup is powered by a 5V power source, ensuring efficient energy use. The implementation demonstrates how integrating IoT with SCADA enhances process transparency, reduces manual intervention, and enables timely responses to safety hazards. This solution offers a scalable and cost-effective model for industries requiring consistent environmental monitoring and quick incident response.

Keywords— Internet of Things (IoT), Supervisory Control and Data Acquisition (SCADA), Carbon Monoxide Sensor (MQ7), Temperature and Humidity Sensor (DHT11), Global System for Mobile Communications (GSM), Liquid Crystal Display (LCD),

Node Microcontroller Unit (NodeMCU), Combustible Gas Sensor (MQ2), Vibration Sensor

I. INTRODUCTION

Industrial environments often face significant risks due to the presence of hazardous gases, fluctuating temperatures, and structural vibrations that can lead to accidents or equipment failures. Manual monitoring of these parameters is not only inefficient but also prone to delays and errors, which can compromise safety. To address this issue, the proposed system introduces an IoT-based SCADA (Supervisory Control and Data Acquisition) solution focused on real-time monitoring and early-warning alerts. The primary objective of this system is to enhance industrial safety by continuously tracking key parameters like gas levels, temperature, humidity, and vibrations using an integrated sensor network. It employs sensors such as MQ7 for carbon monoxide detection, MQ2 for combustible gas, DHT11 for temperature and humidity, and vibration sensors for mechanical instability. Data from these sensors is transmitted via NodeMCU to the ThingSpeak cloud platform for real-time analysis and visualization. When thresholds are breached, the system automatically activates alarms through buzzers and sends SMS notifications via a GSM module to alert concerned personnel immediately. An LCD module provides local real-time feedback to on-site operators. This system aims to reduce the dependency on manual supervision, enable proactive maintenance, and minimize the risk of industrial accidents. By offering a scalable, low-cost, and automated monitoring solution, the prototype meets modern industry demands for smarter and safer operations in high-risk environments.

Key Contributions of This Work:

Real-time industrial monitoring using IoT-enabled SCADA architecture, Multi-sensor fusion for gas, temperature, vibration, and fire detection, Dual alert system using GSM SMS and buzzer, Combined cloud visualization (ThingSpeak) and local LCD display. Scalable and cost-effective design with future support for AI/ML-based predictive analytics

II. LITERATURE SURVEY

We must examine the entire idea underlying the issue in order to do this examination. Studying the current system is more crucial than creating a new one. We learn what kinds of needs have been met thus far from this study, as well as how to use the newest technologies to make the system more sophisticated and effective than it was before. The research paper analysis that follows is part of our suggested system: According to Fawzy A. Osman [1] and N. S. Rani, S. R. Khatri, and P. K. Agarwal [2], we can construct a SCADA system that can be utilized for data acquisition and supervisory control. The IEEE C37.1 standard is used for this. This system allows the various devices found in an industrial setting to be monitored and controlled. Kim, Y.-G., Shahzad, Aa., and Elgamoudi, A. Using online technology, Singh, P. P., Khosla, P. K., & Mittal, M. [4] directed us to the system that allows us to access the process control library from off-campus. [5] Sudip Phuyala, Diwakar Bistaa, and Jan Izykowskib Aghenta, L. O., & Iqbal, M. T. Ganga, D., and Ramachandran, V. [6]. [7] describes the ways in which automation has become a crucial component of industrial growth. The efficiency of our products will significantly rise if we replace the outdated manual processes with modern, automated technologies. With our system, we can make high-quality products faster than with human control. The new, developing software technologies provide the basis of the majority of automated systems. C. Ganeshan and S. Kumar Singh. [8] Al-Sughair, L., Benammar, M., Al-Kuwari, M., Ramadan, A., Ismael, Y., & Gastli, A. [9] suggested a weather monitoring system that makes use of an Arduino development board. Based on their findings, we can use the internet of things to remotely monitor the local weather. Jeng, S.-L., Chieng, W.-H., and Kao, K.-C. [10] However, the technology is limited to tracking the weather. The system is devoid of any governing component. They made use of many sensors, including light, sound, temperature, and others. Merchan, D. F., Peralta, J. A., Minchala, L. I., Vazquez-Rodas, A., & Astudillo-Salinas, D. [11] Shinde, K. S., and P. H. Bhagat [12] suggested research on the network of wireless sensors. They developed a system for monitoring smart environments using WSN technology. Numerous harmful chemicals, including nitrogen oxides and sulfur dioxide, are emitted into the atmosphere as a result of growing industrialization, a

substantial growth in the population, and an increase in automobiles, all of which contribute to pollution. It could end up endangering human life. Therefore, it must be watched over and managed.

III. EXISTING METHOD

Existing models of IoT-based Industrial SCADA systems typically integrate sensors such as MQ series for gas detection, DHT11 for monitoring environmental parameters like temperature and humidity, and vibration sensors to detect equipment malfunctions or security threats. These systems use microcontrollers like Arduino or NodeMCU to collect data the sensors, which is then transmitted through cloud platforms like ThingSpeak for real-time monitoring. GSM modules are employed to send SMS alerts in case of abnormal conditions, such as gas leaks, temperature surges, or unusual vibrations. While these systems offer basic monitoring and alerting capabilities, the existing models may lack advanced analytics or seamless integration with various industrial processes for predictive maintenance and more proactive decision-making. The proposed IoT-based SCADA system is designed to continuously monitor critical environmental and operational parameters using a variety of sensors. The sensor array includes the MQ7 sensor for detecting carbon monoxide (CO), the MQ2 sensor for combustible gas detection, the DHT11 sensor for measuring temperature and humidity, and vibration sensors for monitoring mechanical movement or instability in equipment. These sensors are interfaced with a NodeMCU microcontroller, which serves as the data acquisition and transmission unit. The NodeMCU reads sensor values and transmits the data wirelessly to the ThingSpeak cloud platform over a Wi-Fi connection. ThingSpeak allows for real-time data visualization and remote monitoring through graphical dashboards. In addition to cloud-based visualization, a 16x2 LCD module is connected locally to the NodeMCU to provide on-site, real-time display of sensor readings. This dual-display approach ensures both local and remote visibility of key parameters, enabling operators to respond quickly to any anomalies or changes in the monitored environment.

IV. PROPOSED METHOD

The proposed model for the IoT-based Industrial SCADA system enhances the existing models by incorporating more advanced features for real-time monitoring, predictive analytics, and improved security. The system integrates range of sensors, including MQ7 for carbon monoxide detection, MQ2 for detecting various gases, DHT11 for temperature and humidity measurement, and vibration sensors to monitor abnormal vibrations or potential

equipment failures. A NodeMCU transmits the sensor data to the ThingSpeak cloud platform for real-time analysis and display. Predictive maintenance is made possible by the system's use of advanced analytics to identify patterns and minimize downtime. The system reacts quickly to unusual circumstances by using automation. When sensors detect hazards such as high gas levels, excessive heat, strong vibrations, fire, or intrusions, the GSM module sends SMS alerts to predefined contacts. At the same time, a buzzer is activated for audible on-site warnings. This ensures both local and remote notification, enabling quick action without manual monitoring.

A. Design Overview

The (Fig.1,2,3) IoT-based Industrial SCADA System ensures secure and automated monitoring of industrial parameters like gas levels, temperature, humidity, and vibrations. It collects data using sensors (MQ7, MQ2, DHT11, vibration sensor) and transmits it to ThingSpeak via NodeMCU for real-time analysis. In case of abnormalities, such as high gas levels or temperature, it triggers SMS alerts via a GSM module. The system also activates buzzers and displays data on an LCD for immediate awareness. Powered by a 5V supply, it enhances industrial safety with continuous monitoring and instant alerts.

1. OFF Condition

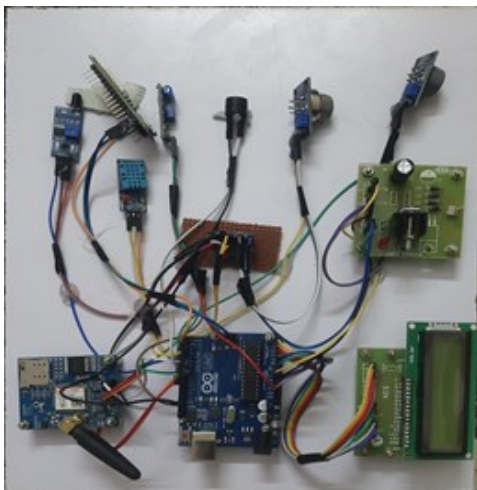


Fig. 1. illustrate the system's transition from OFF state, confirming hardware activation.

2. ON Condition

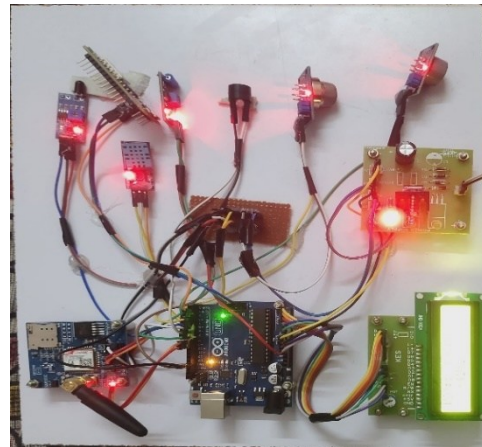


Fig. 2. illustrate the system's transition from OFF state, confirming hardware activation.

B. Flow Diagram

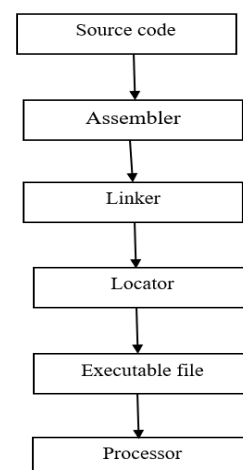


Fig. 3. outlines the operational flow of the SCADA system, from data acquisition to cloud-based monitoring and alert generation.

V. TESTING, RESULTS AND DISCUSSIONS

TESTING

The system was evaluated under controlled conditions. A lighter was used to simulate fire, LPG gas was introduced near the gas sensor, and vibration was manually induced. The temperature sensor was tested using a heat source. Each sensor triggered an automated response, including buzzer activation and SMS alert via GSM module, within 2-3 seconds of detection, demonstrating the system's responsiveness and accuracy.

1. FIRE DETECTION

The Fig.4 shows provided image represents the parameters like analyzing fire and send message to the mobile number provided in the GSM module.

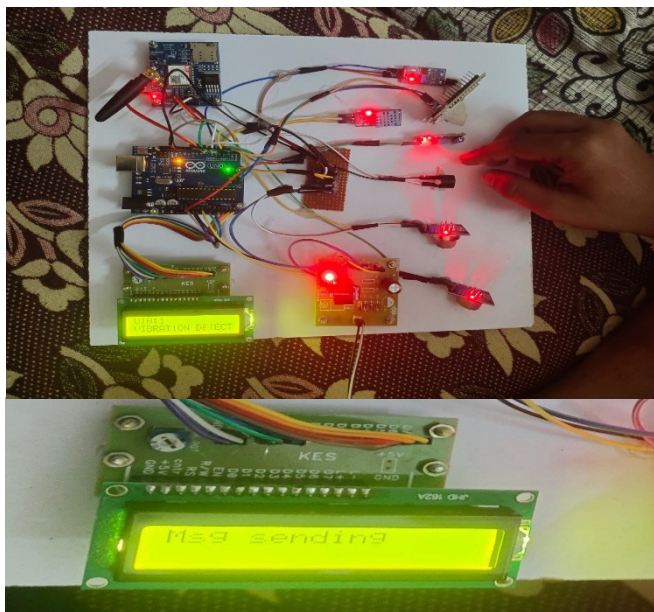


Fig. 4. The flame sensor successfully detected fire presence and activated the GSM and buzzer alert mechanisms.

2. TEMPERATURE DETECTION

The Fig.6 shows provided image represents the current room temperature.

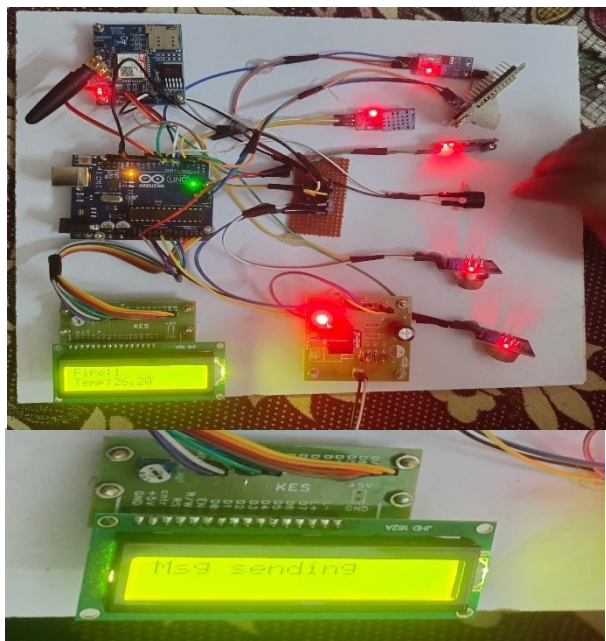


Fig. 5. The DHT11 sensor accurately recorded room temperature, confirming real-time environmental monitoring.

3. VIBRATION DETECTION

The Fig.5 shows provided image represents the parameters for analyzing and sending alert message to provided mobile number when any vibration is detected.

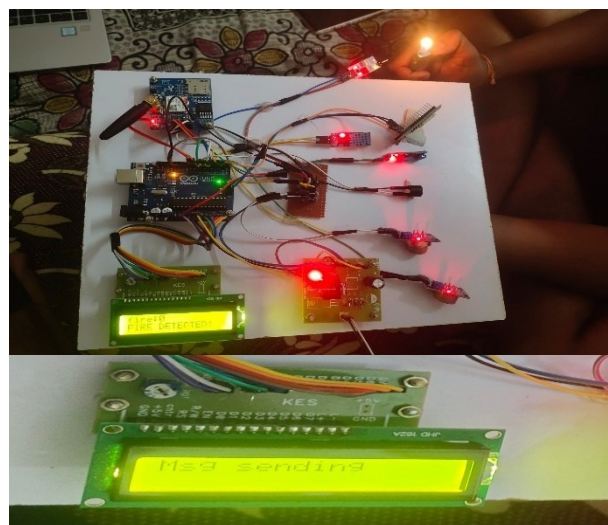


Fig. 6. The vibration sensor detected physical disturbances and triggered alerts, verifying equipment instability response.

4. GAS DETECTION

The Fig.7 shows provided image represents the parameters like analyzing gas and send message alert to the mobile number provided in the GSM module when any gas is detected by gas sensor.

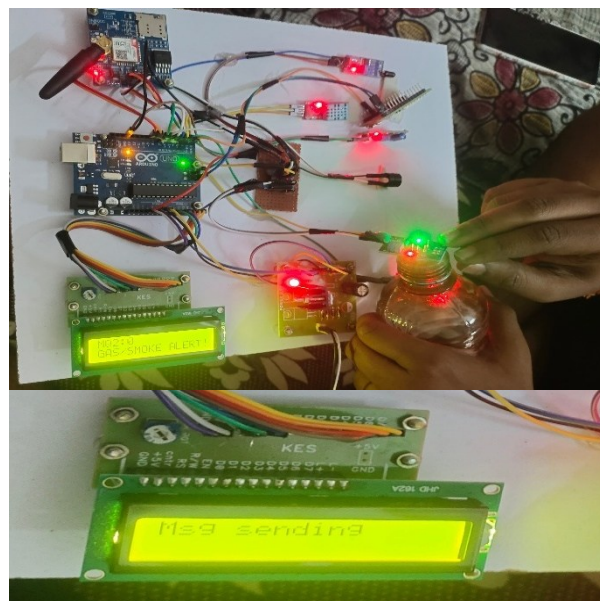


Fig. 7. The gas sensor reacted to combustible gases and triggered the alarm system instantly.

OUTPUTS

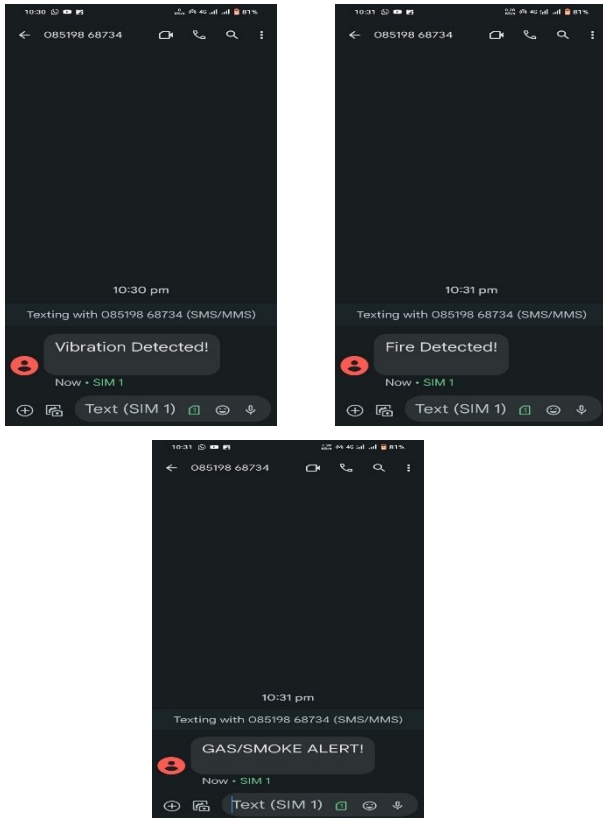


Fig. 8. SMS Alert Received via GSM Module

ANALYSIS

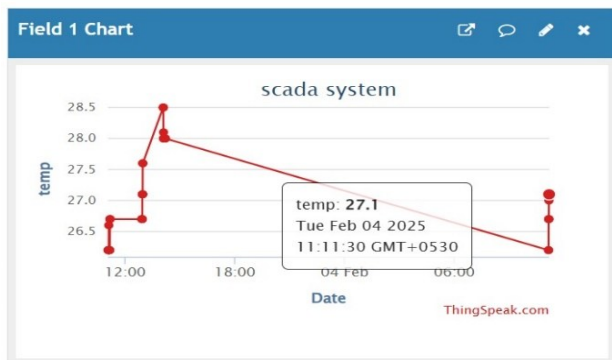


Fig. 9. Shows real-time temperature values. Spikes indicate heat fluctuations.

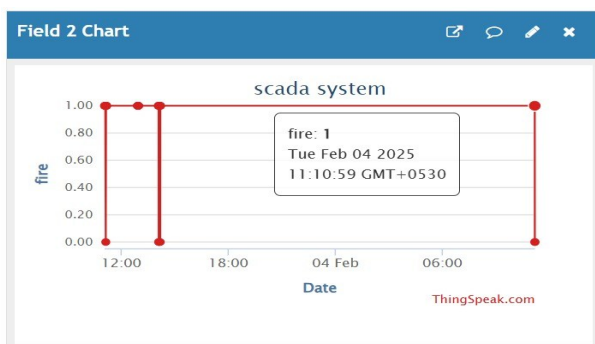


Fig. 10. Fire level graph where peaks signal detected flame presence

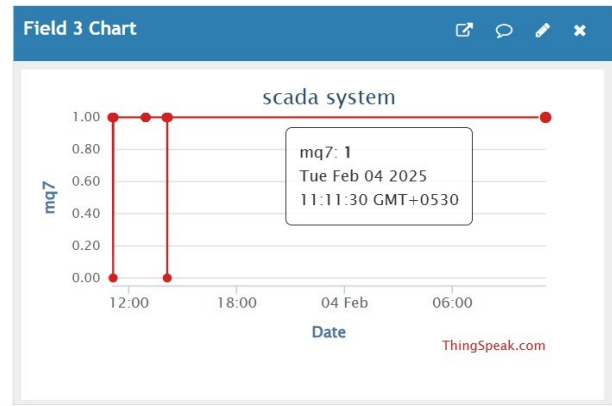


Fig. 11. Gas levels fluctuate based on air quality; spikes mean hazardous gases.

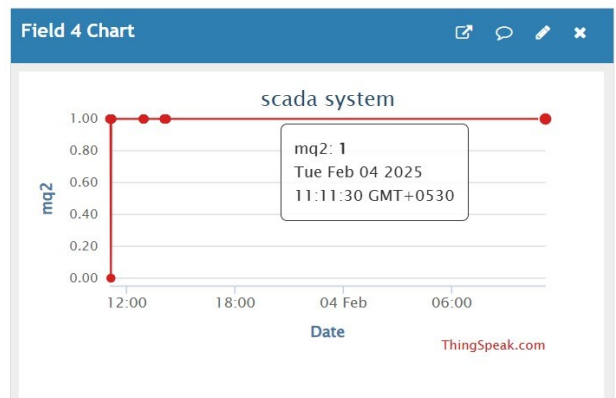


Fig. 12. Carbon monoxide detection curve via MQ7 sensor.

DISCUSSIONS

A Node MCU transmits the sensor data all shown in (Fig.8-12) to the ThingSpeak cloud platform for real-time analysis and display. The system incorporates advanced analytics to detect trends, enabling predictive maintenance and reducing downtime. In case of abnormal conditions such as gas leaks, fire, or security breaches, the system immediately sends SMS alerts through a GSM module and activates buzzers for local notifications. The addition of a fire sensor further enhances safety by detecting fire hazards. The system is powered by a 5V power supply and features an LCD for immediate data display, ensuring high levels of operational safety, efficiency, and automation in industrial environments.

VI. CONCLUSION

An efficient and comprehensive method for real-time monitoring and control of crucial industrial parameters is provided by the suggested IoT-based Industrial SCADA

system. including gas concentration, ambient temperature, humidity, and mechanical vibrations. Utilizing smart sensors such as MQ7, MQ2, DHT11, and vibration sensors, combined with wireless communication through NodeMCU and cloud-based platforms like ThingSpeak, the system ensures timely data collection and live visualization. In the event of abnormal readings, automated alerts are generated via a GSM module, enabling immediate response through SMS notifications and local buzzer alarms. This proactive mechanism significantly improves workplace safety and operational reliability. The system's modular design, remote monitoring capability, and cost efficiency make it suitable for a wide range of industrial settings seeking to adopt smart automation. Overall, the solution enhances situational awareness, minimizes risk, and supports data-driven decision-making in industrial operations.

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